Supplementary Material for Physics-Aware Region Encoding for SAR Text-to-Image Retrieval

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Versioned scope. Sections S1–S4 retain the earlier implementation audits; these are not retrieval evaluations. Section S5 records the spatial-metadata audit and the earlier full-model scale assessment. Sections S6–S8 retain the preceding full-model training, retrieval and exploratory factorial studies. In those historical sections, "PARE" and "full" denote the earlier PARE-F architecture, not the simplified PARE-S model introduced in this revision. Their numbers and negative findings are preserved. Section S9 records the subsequent conditional-token mechanism study; Section S10 specifies the matched PARE-S study and its scale assessment. No earlier model's results are reassigned to the new simplified model. The original test benchmark has already been used during this method-development process.

S1 Historical Extraction Audit

All 919 images in the earlier fixed validation split were processed under five one-factor settings, producing 4,595 per-image records. Every default crop was checked pixel-for-pixel against the then-current unchanged extractor. Table S1 separates mean counts from the percentage of images whose exact unordered bounding-box coordinate set differs from default. A changed box set is not a localization error rate or a pixel-overlap measure. "Clip" is the fraction of selected fine crops whose width or height is less than 24 pixels; the denominator is the number of selected crops for that setting, not the number of images.

Table S1: September extraction diagnostics on 919 earlier-validation images. (η, γ) denote CFAR and CV multipliers; $\bar{n}_c, \bar{n}_f$ are mean crop counts; Δ_c, Δ_f are changed-box-set image percentages. These values describe the old float32 extractor, not the October repair.

(η, γ)	$\bar{n}_c$	$\bar{n}_f$	Δ_c	Δ_f	Clip (%)
$(2.5, 1.3)$	5.546	7.919	0.00	0.00	12.28
$(2.0, 1.3)$	5.546	7.985	0.00	99.56	23.17
$(3.0, 1.3)$	5.546	7.293	0.00	99.56	7.55
$(2.5, 1.1)$	4.243	7.919	99.35	0.00	12.28
$(2.5, 1.5)$	5.817	7.919	99.67	0.00	12.28

At default settings, 895/919 images (97.39%) reached the eight-crop cap; 2/919 images (0.22%) had no fine crop. Similar mean counts therefore did not establish stable spatial support. The median largest coarse bounding box occupied 90.55% of image area, not 90.55% foreground coverage. Of 7,278 default fine crops, 894 (12.28%) were boundary-clipped. These boxes are extraction supports, not semantic annotations. Changing the CFAR multiplier also changes detector fractions in coarse descriptors even when the coarse boxes themselves remain fixed.

Numerical witnesses. On two images, 409 distinct pixel locations became detector-positive as the multiplier increased across the probes, contrary to exact arithmetic with nonnegative ring backgrounds. Every affected position had a negative background in the float32 implementation. Direct float64 ring means were checked at 32 witness positions and were all zero; no claim is made that all 409 were individually checked by direct summation. One of the 7,278 default fine crops was centered on an affected position. All images and detections were retained, including the failed initial monotonicity audit. The count does not establish negligible or harmful retrieval impact.

The subsequent float64 repair and deterministic intensity-tie selection change the extraction implementation. The old table must not be relabeled as a new-threshold experiment. Retrieval-level sensitivity still requires new caches, matched training, and evaluation under the repaired protocol.

S2 October Paired Implementation Check

Table S2: Paired implementation audit on 919 earlier-validation images. Moved unchanged from the October main manuscript. This is not retrieval accuracy and does not use the new clean validation split.

Diagnostic	Earlier	Repaired
Negative ring backgrounds (pixels)	7,235	0
Threshold inversions (pixels)	409	0
Zero-intensity detections (default)	277	0
Selected fine crops	7,278	7,275
Boundary-clipped fine crops (%)	12.28	9.15

Both extractor versions were run on all 919 earlier-validation IDs. Negative ring backgrounds decreased from 7,235 positions on 14 images to zero. Threshold-inversion pixels across multipliers 2.0, 2.5, and 3.0 decreased from 409 to zero, and default zero-intensity detections from 277 to zero. The nonnegative background constraint prevents negative-background detection by design; these counts are not a sensor-calibrated false-alarm estimate.

Direct float64 calculations checked 11,028 local windows and 10,031 rings. Maximum absolute errors were 0 for local means, 4.239×10^{-7} for population standard deviations, and 1.111×10^{-16} for ring means. All repaired crops were pixel-checked against their source. Exact coarse-box sets changed on 467/919 images (50.82%) and fine-box sets on 827/919 (89.99%). Precision-only repair with the old tie rule changed fine sets on seven images; the new tie rule with the old detector changed them on 827 images. These one-factor implementation comparisons are not retrained model ablations.

The number of selected fine crops changed from 7,278 to 7,275, and clipped crops from 894 (12.28%) to 666 (9.15%). A smaller clipping percentage is not evidence of improved retrieval or semantic coverage. The audit preserves the same image list to compare implementations; its earlier fit/validation split had separately identified exact-content duplicates. It therefore does not certify an independent clean validation experiment. New retrieval caches and heads must use the revised content-disjoint protocol.

S3 Historical Training-Pair Simulation

The original training CSV contained 31,968 caption–image rows over 9,191 image IDs, including 29 duplicate exact pairs. An isolated Python random.Random(0) stream simulated the trainer's repeated in-place shuffle for 10 epochs with batch size 64. Each epoch retained its final batch of 32. This is an independent simulation, not a replay of recovered training batches; model initialization can consume the trainer's global random stream.

Of 5,000 simulated batches, 780 (15.6%) contained a repeated image ID. Of 319,680 query exposures, 1,674 (0.524%) had a same-image off-diagonal candidate. These denominators answer different questions and neither quantity measures a retrieval failure rate. They refer to the earlier training population, not to the new content-disjoint split.

For deterministic scores of identical cached documents, let m_i be the number of candidate columns for query i's target image. Grouping all same-image columns in the positive numerator yields

$$\ell_i^{\text{diag}} = \ell_i^{\text{group}} + \log m_i \geq \log m_i. \tag{S1}$$

The repeated columns have identical scores and derivatives with respect to the shared parameters. Thus the loss difference is constant and the two objectives have identical gradients in this setting. The simulated mean $\log m_i$ was 0.00363 nats per query exposure. This quantifies a loss floor; changing only the positive numerator does not remove candidate multiplicity weights in the denominator. Image-unique batching is a different intervention and requires a matched training comparison. Tests check the loss identity, shared-gradient equality, shuffle behavior, and tail-batch retention.

S4 Tensor Storage and Evidence Records

The full head contains $259d + 651$ trainable parameters: 133,259 at $d = 512$ and 199,563 at $d = 768$, checked by instantiating both dimensions. With capacities $M_c = 6$, $M_f = 8$, float32 embeddings and cached statistics, and two int64 counts, the tensor payload per image is

$$4[(M_c + M_f + 2)d + 6(M_c + M_f) + 3] + 16 \text{ bytes.} \tag{S2}$$

This gives 33,132/49,516 bytes for $d = 512/768$. The former was checked against a 16-image encoded smoke cache; the latter is schema-derived. For the earlier 3,688-image gallery, the arithmetic totals are 116.53/174.16 MiB. These exclude metadata strings, serialization, and allocator overhead and are not measurements of runtime or retrieval accuracy. October extraction statistics may be accumulated in float64 before conversion to the unchanged float32 cache schema.

The historical evidence package retains per-image rows, box coordinates, failed and successful audit manifests, numerical witness records, per-batch simulation records, cache summaries, source hashes, and tests. The relevant result directories are `primitive_stability_20260916`, `extractor_fix_20261007`, `training_pair_audit_20260916`, and `cache_budget_20260916`. These local names are provenance identifiers. The October implementation release is available at https://github.com/zhaiji0727/PARE at the October commit `910d74d`; it does not supply the missing historical retrieval checkpoints. Section S6 distinguishes the newly trained November artifacts from that historical record.

The additional uniform-grid control uses 2×3 coarse and 2×4 fine partitions of the full scene, giving six and eight tiles. Its matched crop budget and fusion-head architecture do not match the detector's 24×24 fine-patch spatial scale. The November results below measure its retrieval performance, but this comparison does not isolate spatial-scale effects. Its reliability and token augmentation are disabled. Revised final heads and controls use the learning-rate/contrastive-temperature pair chosen by the main manuscript's fit/validation grid; no historical winning pair is assumed to be optimal for the repaired pipeline.

S5 Spatial-Metadata Audit of the Actual Input Images

We inspected file headers for all 12,869 images in the retained training and official test sets, without decoding or modifying their pixel arrays. Filename extensions did not reliably identify the format: 4,517 files ending in `.png` are GeoTIFFs. These files contain ModelPixelScaleTag (33550), ModelTiepointTag (33922), and GeoKeyDirectoryTag (34735), with a projected model type (GeoKey 1024 = 1) and metre units (GeoKey 3076 = 9001). The tags report horizontal and vertical map-grid spacings from 0.1499977023 to 0.6000324201 m/pixel. These are embedded spacings of the released rasters, not a verification of the original sensor resolution or of all upstream resampling. The remaining 8,352 files are PNGs without additional metadata chunks establishing ground spacing. There were no header-reading errors. GeoTIFF tag semantics are specified in https://docs.ogc.org/is/19-008r4/19-008r4.pdf.

Table S3: Coverage of embedded metre-based projected raster spacing.

Split	All images	Spacing available	Spacing unavailable
Retained training	9,181	3,331	5,850
Test	3,688	1,186	2,502

For observational query groups, both axis spacings are independently rounded to 0.01 m with decimal round-half-up. The rounded values agree for every metadata-bearing test image; thirteen files have raw axis differences no larger than 1.08×10^{-8} m/pixel and round to 0.50 m/pixel on both axes. Groups were defined from metadata before examining retrieval results. The groups contain 760 queries below 0.30 m/pixel, 238 from 0.30 to below 0.45 m/pixel, and 188 at or above 0.45 m/pixel. An additional 2,502 queries have unknown spacing. Evaluation retains the complete 3,688-image gallery for every group; only the query rows are grouped, without re-ranking inside smaller galleries. These comparisons are observational: scene content, geography, sensor and preprocessing may be confounded with spacing. Synthetic resampling is reported separately and is not a new physical SAR acquisition.

Table S4: Prespecified test-query groups and literal country prefixes. File prefixes describe coverage and are not used to infer GSD.

Rounded spacing (m/pixel)	Queries	France	Japan	USA
< 0.30	760	90	45	625
0.30 to < 0.45	238	40	31	167
≥ 0.45	188	18	99	71

Table S5: Full-gallery retrieval metrics (%) by observed stored raster spacing. All queries are ranked against the same complete 3,688-image gallery.

Spacing (m/pixel)	Method	n	R@1	R@5	R@10	MRR@10
< 0.30	Scene-CLIP	760	0.13	0.66	1.32	0.45
< 0.30	Stats-off	760	0.53	2.11	3.55	1.22
< 0.30	PARE	760	1.18	4.61	9.08	2.86
$[0.30, 0.45)$	Scene-CLIP	238	0.00	0.00	0.84	0.12
$[0.30, 0.45)$	Stats-off	238	0.42	2.52	3.78	1.40
$[0.30, 0.45)$	PARE	238	0.84	2.52	7.14	1.93
≥ 0.45	Scene-CLIP	188	3.19	5.85	7.98	4.22
≥ 0.45	Stats-off	188	1.60	3.72	6.91	2.79
≥ 0.45	PARE	188	3.72	11.70	18.62	7.40
Unavailable	Scene-CLIP	2502	0.28	0.92	1.60	0.59
Unavailable	Stats-off	2502	0.36	1.80	3.24	1.05
Unavailable	PARE	2502	0.84	5.00	9.91	2.85

Both stored axes are rounded to 0.01 m before binning. These observational subgroups differ in geography and scene composition; they do not constitute a controlled acquisition-GSD comparison.

S6 Preceding Single-Seed Protocol and Reproducibility

This section documents the preceding study, before the matched five-seed comparison in Section S10. All 20 extraction caches, 45 validation trials, and 20 final heads completed under the protocol below. All five independent evaluations and the three ViT-B/32 scale conditions passed their recorded artifact and rank checks.

Frozen backbones. OpenCLIP uses `ViT-B-32` with `laion2b_s34b_b79k` and `ViT-B-16` with `laion2b_s34b_b88k`. RemoteCLIP uses `ViT-B-32` with `RemoteCLIP-ViT-B-32.pt`; GeoRSCLIP uses `ViT-B-32` with `RS5M_ViT-B-32.pt`. SigLIP uses `ViT-B-16-SigLIP` with `webli`. The shared embedding dimension is 512 for all but SigLIP, which uses 768. External checkpoints are loaded strictly. Each experiment records the actual weight-file hashes, preventing a model name alone from standing in for the weights that were used.

The revised experiment regenerates the physics-guided and uniform-grid caches separately for each of five frozen backbones. The cache manifest binds the image identities, source-file hashes, model and extraction sources, preprocessing, and relevant package versions. A verified import records the original cache execution and its hashes when the cache is used by the optimized training workspace; it does not reattribute cache generation to a different implementation.

The head equations, loss, data order, and initialization protocol are unchanged by the execution optimization. Batched query scoring invokes the same fusion function through PyTorch vectorization. Before formal runs, comparisons of direct and batched scoring checked forward values and parameter gradients in both embedding dimensions, all control modes, and empty-crop cases. Real-image small-run comparisons also checked candidate pools, per-query ranks, and all reported metrics against the reference evaluator. These checks establish implementation agreement; they are not substitutes for the full-data results below.

Each of the 45 validation trials uses 8,263 fit images and 28,759 caption pairs for 10 epochs, and 918 first-caption queries against the complete 918-image validation gallery. The selected learning-rate/temperature pair is then reused by all four final controls for that backbone. Final training uses 9,181 images and 31,945 caption pairs for 10 epochs. No test metric is used for parameter selection. All runs use seed 0, so the results do not estimate variability over independent training seeds.

The external evaluator records its own source and input hashes, ordered query and gallery identities, candidate-pool membership, float32 score matrices, and per-query ranks. It verifies that the four heads match the intended training split, cache contracts, ablation flags, and selected parameters. Summary metrics are independently recomputed from ranks. The candidate-size sweep reuses the complete query–gallery scores; its shared computation is not a measurement of deployment latency at each candidate size. Tests that use 16 images are explicitly excluded from paper results.

The synthetic scale assessment keeps the original-scale ViT-B/32 heads fixed. Each image is downsampled independently by factors 1.5 and 2 using Pillow LANCZOS antialiasing. Target dimensions are the original dimensions divided by the factor and rounded half-up to integers; the image is not enlarged back. All fixed-pixel extraction operations are repeated before the backbone's normal preprocessing. The original caption and target identity are retained, and all 3,688 images remain in each gallery. The transformation signature is recorded separately from the original cache contract. This probes raster-scale perturbation, not a physical change of sensor, speckle, incidence angle, or point-spread function.

Initialization and operating scope. Initial values are $\beta = e$, $s_\rho = \tau = 0.05$, $\rho_{\min} = \mathrm{sigm}(-2)$, $\lambda_{\mathrm{rel}} = 1$, $a = 0.1$, and $u_k \sim \mathcal{N}(0, 0.02^2 I)$; the affine layers use PyTorch Linear defaults. This preceding test experiment uses one prescribed training seed, and does not estimate run-to-run uncertainty. The follow-up in Section S8 uses three seeds on validation only. Query-conditioned fused embeddings are not one fixed gallery embedding for exact cosine search.

Execution environment and availability. The experiment used NVIDIA RTX PRO 6000 Blackwell Server Edition GPUs (97,887 MiB reported memory), driver 590.48.01, Python 3.10.12, PyTorch 2.9.1+cu128, CUDA 12.8, OpenCLIP 2.26.1, Pillow 10.4.0, and NumPy 1.26.4. The independent evaluator source SHA-256 is `9b08150a2803`…; the vectorized-scoring helper is `80661378e018`…. Full hashes, training/evaluation commands, all final heads, split identities, and per-query ranks accompany the public release. Caption text, source imagery, and third-party backbone weights are obtained from their original providers. Published result projections explicitly bind their hashes to the original complete local records; they are not substituted for those originals when calculating paper metrics.

The public release is commit `6e4ed9a` at https://github.com/zhaiji0727/PARE. Final heads are in `checkpoints/revision_20261103/`; numeric summaries and compressed per-query evidence are in

`benchmarks/revision_20261103/`. `docs/NOVEMBER_REPRODUCTION.md` gives the staged commands and exact run labels. The October commit remains accessible in Git history.

The synthetic coarsening and observational subgroup results in Section S5 and Table S28 use the same original-scale ranking population as the preceding single-seed evaluation. All 3,688 original-scale ranks for PARE, statistics-off, and Scene-CLIP exactly match the corresponding separate evaluator for that preceding single-seed study. At 2×, PARE-F R@10 drops by 2.60 percentage points, although it remains above both controls. The metadata subgroups are geographically confounded; their higher recall in the coarsest group is not a causal effect of GSD.

Operating scope. Query-independent extraction can omit local detail; a reliability floor cannot recover missing crops. Backbone preprocessing confounds statistical and spatial-scale effects in grid comparisons. Because the fused image representation depends on the query, it is not one fixed gallery embedding for exact cosine search.

S7 Complete Preceding Single-Seed Retrieval Results

The tables in this section use the preceding November single-seed models; the new five-seed results appear in Section S10. Natural-shortlist misses count as failures. Controlled and hard pools insert the relevant image and therefore answer a different question. R@1/5/10 and MRR@10 are reported as percentages; MRR@10 assigns zero beyond rank 10. Single-seed point estimates do not imply statistical significance.

The strict CFAR-free control exceeds full PARE in full-gallery R@10 and MRR@10 on all five backbones. This control retains soft reliability and radiometric tokens, while replacing the fine attended vector by the scene vector and retaining both gate outputs, excluding fine embeddings from the gate aggregate, and zeroing coarse detector fractions. This combined intervention does not isolate one pathway or compare soft against hard reliability. This unfavorable component result is retained explicitly in the historical single-seed tables below; the revised main Table I reports the later five-seed study.

Table S6: OpenCLIP ViT-B/32: all prespecified natural candidate sizes. There are 3,688 first-caption queries; n_+ counts included positives. Misses count as failures in end-to-end metrics. At $K = 3,688$, all scores are full-gallery metrics. A dash denotes an undefined conditional metric when no positive is included.

				End-to-end (%)				Conditional on inclusion (%)			
Method	K	n_+	Cand. recall	R@1	R@5	R@10	MRR@10	R@1	R@5	R@10	MRR@10
PARE	100	403	10.93	1.08	3.82	5.56	2.28	9.93	34.99	50.87	20.87
	200	653	17.71	1.03	4.28	7.02	2.50	5.82	24.20	39.66	14.14
	500	1174	31.83	1.03	4.80	8.38	2.77	3.24	15.08	26.32	8.69
	1000	1760	47.72	1.06	4.93	8.87	2.88	2.22	10.34	18.58	6.04
	2000	2617	70.96	1.14	5.40	9.60	3.07	1.60	7.60	13.53	4.32
	3688	3688	100.00	1.06	5.10	10.01	3.03	1.06	5.10	10.01	3.03
Stats-off	100	403	10.93	0.38	1.93	3.15	1.07	3.47	17.62	28.78	9.77
	200	653	17.71	0.38	1.93	3.42	1.11	2.14	10.87	19.30	6.27
	500	1174	31.83	0.46	1.98	3.58	1.20	1.45	6.22	11.24	3.78
	1000	1760	47.72	0.46	2.01	3.50	1.20	0.97	4.20	7.33	2.50
	2000	2617	70.96	0.46	2.03	3.52	1.20	0.65	2.87	4.97	1.69
	3688	3688	100.00	0.46	2.01	3.52	1.20	0.46	2.01	3.52	1.20
CFAR-strict	100	403	10.93	1.14	4.12	5.97	2.39	10.42	37.72	54.59	21.83
	200	653	17.71	1.33	4.37	7.16	2.70	7.50	24.66	40.43	15.26
	500	1174	31.83	1.44	4.85	8.46	3.09	4.51	15.25	26.58	9.72
	1000	1760	47.72	1.41	5.26	9.08	3.26	2.95	11.02	19.03	6.83
	2000	2617	70.96	1.44	5.61	10.17	3.45	2.03	7.91	14.33	4.86
	3688	3688	100.00	1.38	5.80	10.39	3.43	1.38	5.80	10.39	3.43
Uniform grid	100	403	10.93	0.22	1.03	2.01	0.63	1.99	9.43	18.36	5.77
	200	653	17.71	0.22	1.03	2.01	0.63	1.23	5.82	11.33	3.56
	500	1174	31.83	0.22	1.03	2.01	0.63	0.68	3.24	6.30	1.98
	1000	1760	47.72	0.22	1.03	2.01	0.63	0.45	2.16	4.20	1.32
	2000	2617	70.96	0.22	1.03	2.01	0.63	0.31	1.45	2.83	0.89
	3688	3688	100.00	0.22	1.03	2.01	0.63	0.22	1.03	2.01	0.63

Table S7: OpenCLIP ViT-B/16: all prespecified natural candidate sizes. There are 3,688 first-caption queries; n_+ counts included positives. Misses count as failures in end-to-end metrics. At $K = 3,688$, all scores are full-gallery metrics. A dash denotes an undefined conditional metric when no positive is included.

				End-to-end (%)				Conditional on inclusion (%)			
Method	K	n_+	Cand. recall	R@1	R@5	R@10	MRR@10	R@1	R@5	R@10	MRR@10
PARE	100	416	11.28	0.89	3.58	5.37	2.06	7.93	31.73	47.60	18.24
	200	646	17.52	0.92	3.82	6.10	2.21	5.26	21.83	34.83	12.61
	500	1124	30.48	1.00	4.07	7.24	2.46	3.29	13.35	23.75	8.07
	1000	1628	44.14	1.03	4.31	7.10	2.48	2.33	9.77	16.09	5.62
	2000	2460	66.70	1.25	4.91	8.35	2.85	1.87	7.36	12.52	4.28
	3688	3688	100.00	1.27	5.07	8.95	2.97	1.27	5.07	8.95	2.97
Stats-off	100	416	11.28	0.46	1.38	2.66	0.93	4.09	12.26	23.56	8.22
	200	646	17.52	0.43	1.41	2.74	0.93	2.48	8.05	15.63	5.31
	500	1124	30.48	0.43	1.41	2.71	0.92	1.42	4.63	8.90	3.03
	1000	1628	44.14	0.43	1.38	2.71	0.92	0.98	3.13	6.14	2.08
	2000	2460	66.70	0.43	1.38	2.71	0.92	0.65	2.07	4.07	1.38
	3688	3688	100.00	0.43	1.38	2.68	0.92	0.43	1.38	2.68	0.92
CFAR-strict	100	416	11.28	1.49	4.20	5.67	2.58	13.22	37.26	50.24	22.91
	200	646	17.52	1.36	4.20	6.86	2.72	7.74	23.99	39.16	15.53
	500	1124	30.48	1.41	4.91	8.13	3.02	4.63	16.10	26.69	9.91
	1000	1628	44.14	1.36	5.07	8.51	3.09	3.07	11.49	19.29	6.99
	2000	2460	66.70	1.41	5.78	9.52	3.43	2.11	8.66	14.27	5.14
	3688	3688	100.00	1.38	5.91	10.03	3.46	1.38	5.91	10.03	3.46
Uniform grid	100	416	11.28	0.54	2.09	3.36	1.15	4.81	18.51	29.81	10.20
	200	646	17.52	0.54	2.17	3.52	1.18	3.10	12.38	20.12	6.75
	500	1124	30.48	0.54	2.22	3.71	1.21	1.78	7.30	12.19	3.98
	1000	1628	44.14	0.54	2.22	3.66	1.20	1.23	5.04	8.29	2.73
	2000	2460	66.70	0.54	2.22	3.66	1.20	0.81	3.33	5.49	1.81
	3688	3688	100.00	0.54	2.22	3.66	1.20	0.54	2.22	3.66	1.20

Table S8: RemoteCLIP: all prespecified natural candidate sizes. There are 3,688 first-caption queries; n_+ counts included positives. Misses count as failures in end-to-end metrics. At $K = 3,688$, all scores are full-gallery metrics. A dash denotes an undefined conditional metric when no positive is included.

				End-to-end (%)				Conditional on inclusion (%)			
Method	K	n_+	Cand. recall	R@1	R@5	R@10	MRR@10	R@1	R@5	R@10	MRR@10
PARE	100	336	9.11	0.49	1.98	2.90	1.11	5.36	21.73	31.85	12.18
	200	526	14.26	0.46	1.98	3.20	1.11	3.23	13.88	22.43	7.81
	500	1027	27.85	0.43	1.87	3.06	1.06	1.56	6.72	11.00	3.82
	1000	1740	47.18	0.46	2.06	3.39	1.15	0.98	4.37	7.18	2.44
	2000	2697	73.13	0.43	2.03	3.47	1.13	0.59	2.78	4.75	1.55
	3688	3688	100.00	0.41	2.03	3.47	1.11	0.41	2.03	3.47	1.11
Stats-off	100	336	9.11	0.38	1.55	2.77	0.95	4.17	16.96	30.36	10.41
	200	526	14.26	0.35	1.52	2.93	0.95	2.47	10.65	20.53	6.65
	500	1027	27.85	0.35	1.57	3.01	0.96	1.27	5.65	10.81	3.45
	1000	1740	47.18	0.35	1.57	3.01	0.96	0.75	3.33	6.38	2.03
	2000	2697	73.13	0.35	1.60	3.06	0.97	0.48	2.19	4.19	1.32
	3688	3688	100.00	0.35	1.60	3.06	0.97	0.35	1.60	3.06	0.97
CFAR-strict	100	336	9.11	0.68	2.58	3.77	1.46	7.44	28.27	41.37	16.00
	200	526	14.26	0.62	2.44	4.01	1.45	4.37	17.11	28.14	10.17
	500	1027	27.85	0.65	2.28	4.20	1.43	2.34	8.18	15.09	5.12
	1000	1740	47.18	0.65	2.22	4.23	1.45	1.38	4.71	8.97	3.07
	2000	2697	73.13	0.68	2.20	4.39	1.50	0.93	3.00	6.01	2.05
	3688	3688	100.00	0.70	2.33	4.50	1.56	0.70	2.33	4.50	1.56
Uniform grid	100	336	9.11	0.33	1.11	2.14	0.73	3.57	12.20	23.51	7.96
	200	526	14.26	0.33	1.08	2.11	0.72	2.28	7.60	14.83	5.04
	500	1027	27.85	0.33	1.06	2.11	0.71	1.17	3.80	7.59	2.56
	1000	1740	47.18	0.33	1.06	2.09	0.71	0.69	2.24	4.43	1.50
	2000	2697	73.13	0.33	1.06	2.06	0.71	0.44	1.45	2.82	0.97
	3688	3688	100.00	0.33	1.06	2.06	0.71	0.33	1.06	2.06	0.71

Table S9: SigLIP: all prespecified natural candidate sizes. There are 3,688 first-caption queries; n_+ counts included positives. Misses count as failures in end-to-end metrics. At $K = 3,688$, all scores are full-gallery metrics. A dash denotes an undefined conditional metric when no positive is included.

				End-to-end (%)				Conditional on inclusion (%)			
Method	K	n_+	Cand. recall	R@1	R@5	R@10	MRR@10	R@1	R@5	R@10	MRR@10
PARE	100	308	8.35	0.92	2.87	4.07	1.79	11.04	34.42	48.70	21.41
	200	509	13.80	0.87	3.15	4.66	1.89	6.29	22.79	33.79	13.67
	500	899	24.38	0.79	3.44	5.78	1.98	3.23	14.13	23.69	8.14
	1000	1408	38.18	0.98	3.93	6.72	2.31	2.56	10.30	17.61	6.05
	2000	2437	66.08	1.11	4.64	7.86	2.61	1.68	7.02	11.90	3.94
	3688	3688	100.00	1.22	4.91	8.22	2.78	1.22	4.91	8.22	2.78
Stats-off	100	308	8.35	0.60	1.65	2.39	1.05	7.14	19.81	28.57	12.52
	200	509	13.80	0.60	1.65	2.44	1.05	4.32	11.98	17.68	7.58
	500	899	24.38	0.60	1.68	2.55	1.06	2.45	6.90	10.46	4.36
	1000	1408	38.18	0.60	1.71	2.60	1.08	1.56	4.47	6.82	2.82
	2000	2437	66.08	0.60	1.71	2.63	1.08	0.90	2.59	3.98	1.63
	3688	3688	100.00	0.60	1.71	2.63	1.08	0.60	1.71	2.63	1.08
CFAR-strict	100	308	8.35	0.87	3.04	4.34	1.78	10.39	36.36	51.95	21.36
	200	509	13.80	1.19	3.47	5.07	2.14	8.64	25.15	36.74	15.51
	500	899	24.38	1.27	3.88	6.10	2.45	5.23	15.91	25.03	10.04
	1000	1408	38.18	1.38	4.15	7.05	2.68	3.62	10.87	18.47	7.03
	2000	2437	66.08	1.46	5.21	8.92	3.16	2.22	7.88	13.50	4.78
	3688	3688	100.00	1.52	5.45	9.44	3.27	1.52	5.45	9.44	3.27
Uniform grid	100	308	8.35	0.54	1.74	2.63	1.08	6.49	20.78	31.49	12.88
	200	509	13.80	0.52	1.71	2.66	1.05	3.73	12.38	19.25	7.62
	500	899	24.38	0.54	1.79	2.74	1.10	2.22	7.34	11.23	4.51
	1000	1408	38.18	0.54	1.82	2.79	1.11	1.42	4.76	7.32	2.91
	2000	2437	66.08	0.54	1.82	2.79	1.11	0.82	2.75	4.23	1.68
	3688	3688	100.00	0.54	1.82	2.79	1.11	0.54	1.82	2.79	1.11

Table S10: GeoRSCLIP: all prespecified natural candidate sizes. There are 3,688 first-caption queries; n_+ counts included positives. Misses count as failures in end-to-end metrics. At $K = 3,688$, all scores are full-gallery metrics. A dash denotes an undefined conditional metric when no positive is included.

				End-to-end (%)				Conditional on inclusion (%)			
Method	K	n_+	Cand. recall	R@1	R@5	R@10	MRR@10	R@1	R@5	R@10	MRR@10
PARE	100	590	16.00	1.52	4.96	8.00	3.07	9.49	31.02	50.00	19.22
	200	908	24.62	1.63	5.53	9.06	3.36	6.61	22.47	36.78	13.66
	500	1518	41.16	1.57	5.69	9.52	3.42	3.82	13.83	23.12	8.30
	1000	2154	58.41	1.49	5.83	9.82	3.48	2.55	9.98	16.81	5.96
	2000	3002	81.40	1.49	5.91	10.11	3.53	1.83	7.26	12.43	4.34
	3688	3688	100.00	1.49	6.02	10.03	3.56	1.49	6.02	10.03	3.56
Stats-off	100	590	16.00	0.33	1.36	2.90	0.87	2.03	8.47	18.14	5.45
	200	908	24.62	0.33	1.36	2.90	0.87	1.32	5.51	11.78	3.54
	500	1518	41.16	0.33	1.36	2.90	0.87	0.79	3.29	7.05	2.12
	1000	2154	58.41	0.33	1.36	2.90	0.87	0.56	2.32	4.97	1.49
	2000	3002	81.40	0.33	1.36	2.90	0.87	0.40	1.67	3.56	1.07
	3688	3688	100.00	0.33	1.36	2.90	0.87	0.33	1.36	2.90	0.87
CFAR-strict	100	590	16.00	1.60	5.88	8.51	3.42	10.00	36.78	53.22	21.36
	200	908	24.62	1.68	6.48	10.09	3.82	6.83	26.32	40.97	15.51
	500	1518	41.16	1.76	6.97	11.28	4.03	4.28	16.93	27.40	9.79
	1000	2154	58.41	1.84	6.64	11.71	4.09	3.16	11.37	20.06	7.01
	2000	3002	81.40	1.93	6.91	12.20	4.24	2.37	8.49	14.99	5.21
	3688	3688	100.00	1.98	7.08	12.34	4.31	1.98	7.08	12.34	4.31
Uniform grid	100	590	16.00	0.49	2.09	3.77	1.22	3.05	13.05	23.56	7.65
	200	908	24.62	0.49	2.09	3.74	1.22	1.98	8.48	15.20	4.94
	500	1518	41.16	0.49	2.11	3.74	1.22	1.19	5.14	9.09	2.97
	1000	2154	58.41	0.49	2.11	3.74	1.22	0.84	3.62	6.41	2.09
	2000	3002	81.40	0.49	2.11	3.74	1.22	0.60	2.60	4.60	1.50
	3688	3688	100.00	0.49	2.11	3.74	1.22	0.49	2.11	3.74	1.22

Table S11: OpenCLIP ViT-B/32: Positive plus 99 uniformly sampled distractors; five seed-0 pools per query. All 3,688 queries are used. These positive-inserted diagnostics are not end-to-end natural retrieval. All metrics are percentages.

Method	R@1	R@5	R@10	MRR@10
Scene-CLIP	4.92	16.15	26.10	10.06
Mean-pool	2.29	8.92	14.88	5.28
MaxSim	4.77	15.53	24.24	9.57
PARE	21.31	55.95	72.84	35.89
Stats-off	9.68	29.74	43.51	18.40
CFAR-strict	22.86	58.57	74.60	37.87
Uniform grid	5.51	18.82	29.68	11.41

Table S12: OpenCLIP ViT-B/16: Positive plus 99 uniformly sampled distractors; five seed-0 pools per query. All 3,688 queries are used. These positive-inserted diagnostics are not end-to-end natural retrieval. All metrics are percentages.

Method	R@1	R@5	R@10	MRR@10
Scene-CLIP	5.11	16.25	24.93	10.05
Mean-pool	1.79	7.89	14.36	4.60
MaxSim	4.01	12.24	19.25	7.71
PARE	20.17	54.00	72.04	34.61
Stats-off	7.10	23.42	35.72	14.32
CFAR-strict	22.79	58.27	75.21	37.87
Uniform grid	9.08	27.68	40.72	17.16

Table S13: RemoteCLIP: Positive plus 99 uniformly sampled distractors; five seed-0 pools per query. All 3,688 queries are used. These positive-inserted diagnostics are not end-to-end natural retrieval. All metrics are percentages.

Method	R@1	R@5	R@10	MRR@10
Scene-CLIP	3.77	13.38	21.82	8.17
Mean-pool	1.43	6.97	13.00	4.01
MaxSim	3.36	12.22	20.13	7.36
PARE	10.44	35.28	52.71	21.20
Stats-off	8.03	28.42	46.05	17.25
CFAR-strict	12.71	40.95	58.93	24.82
Uniform grid	5.11	17.98	28.98	10.92

Table S14: SigLIP: Positive plus 99 uniformly sampled distractors; five seed-0 pools per query. All 3,688 queries are used. These positive-inserted diagnostics are not end-to-end natural retrieval. All metrics are percentages.

Method	R@1	R@5	R@10	MRR@10
Scene-CLIP	3.80	12.80	19.90	7.78
Mean-pool	2.17	8.76	14.32	5.04
MaxSim	3.87	12.64	19.51	7.72
PARE	19.58	54.96	72.31	34.48
Stats-off	6.52	23.17	36.78	13.86
CFAR-strict	22.17	58.54	75.50	37.56
Uniform grid	7.59	24.86	38.53	15.31

Table S15: GeoRSCLIP: Positive plus 99 uniformly sampled distractors; five seed-0 pools per query. All 3,688 queries are used. These positive-inserted diagnostics are not end-to-end natural retrieval. All metrics are percentages.

Method	R@1	R@5	R@10	MRR@10
Scene-CLIP	7.20	22.81	34.32	14.01
Mean-pool	3.13	11.80	19.15	6.99
MaxSim	5.93	17.05	25.34	10.83
PARE	23.06	60.29	77.26	38.71
Stats-off	7.20	22.81	34.32	14.01
CFAR-strict	25.24	62.47	78.73	40.94
Uniform grid	9.41	28.60	42.19	17.79

Table S16: OpenCLIP ViT-B/32: Positive plus the 49 strongest scene-level negatives. All 3,688 queries are used. These positive-inserted diagnostics are not end-to-end natural retrieval. All metrics are percentages.

Method	R@1	R@5	R@10	MRR@10
Scene-CLIP	0.38	1.06	1.82	0.72
Mean-pool	0.24	1.55	3.58	0.92
MaxSim	0.24	1.38	2.11	0.71
PARE	17.68	46.26	62.99	30.03
Stats-off	2.52	13.23	25.27	7.45
CFAR-strict	18.57	47.40	65.81	31.30
Uniform grid	0.22	1.03	2.06	0.64

Table S17: OpenCLIP ViT-B/16: Positive plus the 49 strongest scene-level negatives. All 3,688 queries are used. These positive-inserted diagnostics are not end-to-end natural retrieval. All metrics are percentages.

Method	R@1	R@5	R@10	MRR@10
Scene-CLIP	0.33	1.22	1.98	0.73
Mean-pool	0.16	1.14	2.66	0.64
MaxSim	0.43	1.49	2.55	0.94
PARE	18.25	45.47	61.04	29.85
Stats-off	0.62	3.47	8.41	2.14
CFAR-strict	22.29	49.65	64.40	34.07
Uniform grid	1.19	5.59	11.82	3.39

Table S18: RemoteCLIP: Positive plus the 49 strongest scene-level negatives. All 3,688 queries are used. These positive-inserted diagnostics are not end-to-end natural retrieval. All metrics are percentages.

Method	R@1	R@5	R@10	MRR@10
Scene-CLIP	0.24	0.89	1.57	0.56
Mean-pool	1.19	4.77	10.11	3.03
MaxSim	0.27	1.17	1.95	0.70
PARE	4.39	17.38	32.43	10.66
Stats-off	1.25	10.41	23.51	5.86
CFAR-strict	5.72	23.70	42.38	14.17
Uniform grid	0.33	1.49	3.82	1.00

Table S19: SigLIP: Positive plus the 49 strongest scene-level negatives. All 3,688 queries are used. These positive-inserted diagnostics are not end-to-end natural retrieval. All metrics are percentages.

Method	R@1	R@5	R@10	MRR@10
Scene-CLIP	0.43	1.19	1.60	0.72
Mean-pool	0.19	1.17	2.47	0.65
MaxSim	0.41	1.19	1.82	0.75
PARE	24.70	55.72	70.55	37.92
Stats-off	1.57	8.49	16.76	4.83
CFAR-strict	26.55	54.91	70.77	38.72
Uniform grid	1.08	5.26	10.87	3.18

Table S20: GeoRSCLIP: Positive plus the 49 strongest scene-level negatives. All 3,688 queries are used. These positive-inserted diagnostics are not end-to-end natural retrieval. All metrics are percentages.

Method	R@1	R@5	R@10	MRR@10
Scene-CLIP	0.33	1.36	2.90	0.87
Mean-pool	0.22	1.27	3.36	0.85
MaxSim	0.38	1.57	3.15	1.00
PARE	17.33	44.47	61.71	29.08
Stats-off	0.33	1.36	2.90	0.87
CFAR-strict	17.71	46.07	64.10	29.94
Uniform grid	0.95	5.83	12.15	3.29

Table S21: OpenCLIP ViT-B/32: all nine 10-epoch validation trials, with 8,263 fit images and 918 first-caption validation queries/gallery images. Selection maximizes R@10, then MRR@10, then retains the first grid trial on an exact tie. Test results are not used for selection.

Trial	Learning rate	κ	Val R@1	Val R@5	Val R@10	Val MRR@10	Selected
1	0.0005	0.03	2.07	9.69	16.34	5.51	
2	0.0005	0.05	1.42	6.86	12.53	3.99	
3	0.0005	0.07	1.63	6.54	11.66	3.96	
4	0.001	0.03	2.51	11.33	19.39	6.47	
5	0.001	0.05	2.07	9.80	16.67	5.46	
6	0.001	0.07	1.85	8.82	15.69	5.09	
7	0.002	0.03	4.36	13.62	23.75	8.68	Yes
8	0.002	0.05	2.18	11.76	20.81	6.41	
9	0.002	0.07	2.18	9.69	17.65	5.72	

Table S22: OpenCLIP ViT-B/16: all nine 10-epoch validation trials, with 8,263 fit images and 918 first-caption validation queries/gallery images. Selection maximizes R@10, then MRR@10, then retains the first grid trial on an exact tie. Test results are not used for selection.

Trial	Learning rate	κ	Val R@1	Val R@5	Val R@10	Val MRR@10	Selected
1	0.0005	0.03	1.85	8.61	16.23	5.28	
2	0.0005	0.05	0.98	7.30	13.18	3.85	
3	0.0005	0.07	0.87	6.64	11.11	3.38	
4	0.001	0.03	2.51	11.87	20.48	6.91	
5	0.001	0.05	1.85	10.68	19.06	5.83	
6	0.001	0.07	2.51	8.61	14.38	5.24	
7	0.002	0.03	2.51	12.75	22.88	7.44	
8	0.002	0.05	2.83	14.16	23.42	7.66	Yes
9	0.002	0.07	2.51	10.24	18.74	6.18	

Table S23: RemoteCLIP: all nine 10-epoch validation trials, with 8,263 fit images and 918 first-caption validation queries/gallery images. Selection maximizes R@10, then MRR@10, then retains the first grid trial on an exact tie. Test results are not used for selection.

Trial	Learning rate	κ	Val R@1	Val R@5	Val R@10	Val MRR@10	Selected
1	0.0005	0.03	1.42	5.45	8.93	3.17	
2	0.0005	0.05	1.20	5.23	8.82	3.16	
3	0.0005	0.07	0.54	4.25	7.52	2.11	
4	0.001	0.03	1.63	6.10	9.80	3.66	
5	0.001	0.05	1.31	5.12	9.37	2.98	
6	0.001	0.07	0.87	5.45	9.69	2.85	
7	0.002	0.03	1.42	7.19	12.20	4.04	
8	0.002	0.05	1.42	6.86	12.53	3.96	Yes
9	0.002	0.07	1.42	6.54	11.33	3.75	

Table S24: SigLIP: all nine 10-epoch validation trials, with 8,263 fit images and 918 first-caption validation queries/gallery images. Selection maximizes R@10, then MRR@10, then retains the first grid trial on an exact tie. Test results are not used for selection.

Trial	Learning rate	κ	Val R@1	Val R@5	Val R@10	Val MRR@10	Selected
1	0.0005	0.03	1.85	8.06	14.60	4.86	
2	0.0005	0.05	1.20	5.66	11.00	3.34	
3	0.0005	0.07	0.65	3.59	6.97	2.05	
4	0.001	0.03	2.72	11.55	19.72	6.65	
5	0.001	0.05	2.40	10.68	18.41	6.12	
6	0.001	0.07	1.85	8.50	15.25	5.05	
7	0.002	0.03	2.40	13.40	22.22	7.27	
8	0.002	0.05	2.72	12.75	21.90	7.12	
9	0.002	0.07	3.27	12.75	22.33	7.46	Yes

Table S25: GeoRSCLIP: all nine 10-epoch validation trials, with 8,263 fit images and 918 first-caption validation queries/gallery images. Selection maximizes R@10, then MRR@10, then retains the first grid trial on an exact tie. Test results are not used for selection.

Trial	Learning rate	κ	Val R@1	Val R@5	Val R@10	Val MRR@10	Selected
1	0.0005	0.03	2.40	8.17	13.18	4.94	
2	0.0005	0.05	2.18	8.06	14.27	4.86	
3	0.0005	0.07	2.07	7.73	13.29	4.56	
4	0.001	0.03	2.40	9.15	15.80	5.62	
5	0.001	0.05	4.14	13.29	20.81	7.87	
6	0.001	0.07	3.38	12.20	20.59	7.29	
7	0.002	0.03	3.49	12.31	21.24	7.55	
8	0.002	0.05	5.01	16.23	26.36	9.97	Yes
9	0.002	0.07	3.81	15.03	24.07	8.68	

Table S26: Validation-selected parameters and validation metrics (percent). Each backbone's full head and controls use the same selected settings for final retraining.

Backbone	Learning rate	κ	Val R@1	Val R@5	Val R@10	Val MRR@10
OpenCLIP ViT-B/32	0.002	0.03	4.36	13.62	23.75	8.68
OpenCLIP ViT-B/16	0.002	0.05	2.83	14.16	23.42	7.66
RemoteCLIP	0.002	0.05	1.42	6.86	12.53	3.96
SigLIP	0.002	0.07	3.27	12.75	22.33	7.46
GeoRSCLIP	0.002	0.05	5.01	16.23	26.36	9.97

Table S27: Signed full-gallery differences in percentage points, computed from unrounded metrics. Negative differences are retained.

Backbone	Comparison	Δ R@1	Δ R@5	Δ R@10	Δ MRR@10
OpenCLIP ViT-B/32	PARE minus Stats-off	+0.60	+3.09	+6.48	+1.83
OpenCLIP ViT-B/32	PARE minus Scene-CLIP	+0.68	+4.04	+8.19	+2.31
OpenCLIP ViT-B/16	PARE minus Stats-off	+0.84	+3.69	+6.26	+2.06
OpenCLIP ViT-B/16	PARE minus Scene-CLIP	+0.95	+3.85	+6.97	+2.25
RemoteCLIP	PARE minus Stats-off	+0.05	+0.43	+0.41	+0.15
RemoteCLIP	PARE minus Scene-CLIP	+0.16	+1.14	+1.90	+0.55
SigLIP	PARE minus Stats-off	+0.62	+3.20	+5.59	+1.70
SigLIP	PARE minus Scene-CLIP	+0.79	+3.71	+6.62	+2.06
GeoRSCLIP	PARE minus Stats-off	+1.17	+4.66	+7.13	+2.68
GeoRSCLIP	PARE minus Scene-CLIP	+1.17	+4.66	+7.13	+2.68

Table S28: OpenCLIP ViT-B/32: complete synthetic scale sensitivity, using the same original-scale heads and all 3,688 queries/gallery images. All four metrics are percentages; changes are relative to original spacing. No physical acquisition-GSD invariance is inferred.

Method	Spacing	R@1	R@5	R@10	MRR@10	Δ R@10 (pp)
Scene-CLIP	Original	0.38	1.06	1.82	0.72	0.00
	1.5x	0.38	1.14	1.93	0.72	+0.11
	2x	0.41	1.14	1.84	0.74	+0.03
PARE	Original	1.06	5.10	10.01	3.03	0.00
	1.5x	1.36	4.56	8.41	2.88	-1.60
	2x	0.87	4.07	7.40	2.29	-2.60
Stats-off	Original	0.46	2.01	3.52	1.20	0.00
	1.5x	0.52	1.79	3.23	1.15	-0.30
	2x	0.38	1.57	3.17	0.99	-0.35

Table S29: Fixed factor definitions. The off state changes only the named pathway; the remaining base parameters, masks, normalization, and scene fallbacks are retained.

Factor	State 1	State 0
A	Original fine attended vector v^f.	Set $v^f = e^g$ while retaining the fine gate output g_f.
B	Unaugmented gate mean over scene, valid coarse, and valid fine embeddings.	Use $(e^g + \sum_{r \in R^c} e^c_r)/(1 + \lvert R^c \rvert)$, without renormalization.
C	Original detector-positive fraction in each coarse descriptor.	Set only that fourth coarse-descriptor entry to zero; retain fine descriptors and all other statistics.
R	Original learned soft reliability in log-attention.	Use constant reliability on valid crops, retaining the original valid mask and constant log-attention term.
T	Original descriptor MLP, three learned prototypes, and token scale.	Normalize the original crop embedding without token augmentation; reliability is unchanged.

S8 Exploratory Factorial Follow-up on the Validation Set

Purpose and evidence status. The five-backbone test results motivated this follow-up: the strict CFAR-free control changes three pathways together, so its comparison with full PARE does not isolate any one of them. We fixed the configurations, seeds, training recipe, contrasts, and reporting rules before running the follow-up. The hypothesis was nevertheless post hoc, and the 918-image validation set had already supported hyperparameter selection. This is therefore an exploratory validation analysis, not a new blinded test. The original full PARE definition, five-backbone test results, and spatial-scale results are retained irrespective of these outcomes.

Factors and matched training. Table S29 defines five binary interventions. All eight ABC combinations use $R = T = 1$. At each of $ABC = 000$ and $ABC = 111$, we additionally evaluate $RT = 00, 01, 10$; the $RT = 11$ endpoint is shared with the eight-cell design. This gives 14 configurations, each trained with seeds 0, 1, and 2, for 42 runs. The full endpoint is $ABC111, R1T1$; the strict endpoint is $ABC000, R1T1$. The legacy fine-branch-only ablation corresponds to $ABC011, R1T1$.

When $A = 0$, the scene coefficient is $g_f + [1 - g_c - g_f]_+$, not generally $1 - g_c$. Turning B off does not disable the fine attended path when $A = 1$. Turning C off does not change fine descriptors, μ, t, ν, crop embeddings, or extraction. Turning both R and T off still retains the same physically motivated crop extraction. These distinctions prevent an intervention from being mistaken for removal of every source of fine-scale or statistical evidence.

All runs use frozen OpenCLIP ViT-B/32 with `laion2b_s34b_b79k` and the unchanged physics training cache. Training uses only the 28,759 caption–image pairs associated with 8,263 fit images; the 918 validation images contribute no loss terms. The fixed recipe is AdamW, learning rate 0.002, contrastive temperature 0.03, weight decay 10^{-4}, batch size 64, and 10 complete epochs in float32 without AMP or early stopping. Each epoch contains 450 batches, including a final batch of 23 pairs. The learning rate and temperature are the previously selected ViT-B/32 settings; there is no new sweep. For each seed, all configurations start with identical base parameters, initial caption-pair order, and per-epoch shuffle sequence. The original one-direction diagonal contrastive loss and duplicate-image behavior are unchanged. Heads previously trained on all 9,181 retained training images are not used in this validation comparison.

Evaluation, contrasts, and verification. Each final-epoch head scores all 918 first-caption validation queries against all 918 validation images, without candidate truncation or positive insertion. Query order follows first occurrence in the clean caption loader, gallery order follows the original cache rows, and the reference descending `torch.argsort` tie handling is retained. R@10 is the primary descriptive metric; R@1, R@5, and MRR@10 are also reported, with reciprocal rank set to zero beyond rank 10. Every contrast is evaluated within each seed before taking its arithmetic mean and sample standard deviation ($n = 3$, denominator $n - 1$). The fixed 36 contrasts comprise 12 conditional ABC effects, three average effects, three pairwise differences-in-differences averaged over the remaining factor, one three-way difference, the full-minus-strict contrast, and eight RT contrasts at each endpoint. Nine were designated primary: $A \mid B = C = 1$, $B \mid A = C = 1$, $C \mid A = B = 1$, and, at each endpoint, $R \mid T = 1$, $T \mid R = 1$, and $RT11 - RT00$. No coefficients are rescaled or selected according to their outcomes.

All 42 saved checkpoints and 918×918 score matrices were read back and checked for configuration, population, source/input hashes, finite values, paired initialization, and shuffle identities. Replaying the saved matrices on the reference CUDA sorting backend reproduced every stored query rank and all four

reported metrics. The original feature-cache hash remained unchanged. A separate anchor check compared the seed-0 full endpoint with the earlier fit-only hyperparameter trial at the same learning rate and temperature. All 15 parameter tensors were bitwise equal (maximum absolute error zero), and all 918 ranks and four metrics agreed. This is equality of learned tensors and results, not equality of the serialized checkpoint files, which carry different metadata. It checks continuity with the prior fit-only run, not independent test generalization.

Results and scope. Figure S1 shows every primary R@10 contrast in the protocol order. With reliability enabled, token augmentation increases R@10 by 17.393 ± 3.587 percentage points at the strict endpoint and 12.200 ± 1.284 points at the full endpoint; all three paired-seed values are positive in both cases. The joint $RT11 - RT00$ changes are 17.792 ± 2.948 and 12.309 ± 1.571 points, respectively. In contrast, reliability with tokens enabled changes R@10 by -2.578 ± 2.903 points at the strict endpoint and 0.036 ± 1.309 points at the full endpoint. The strict-endpoint effects are negative for all three seeds, whereas the full-endpoint effects change sign. Thus the evidence favors the latent token path under these tested settings but does not show a consistent reliability benefit. Tokens remain retrieval-learned features, not identified scattering mechanisms.

At the full ABC context, the fine attended-vector effect $A \mid B = C = 1$ is 0.182 ± 0.926 points, with paired-seed values $0.545, -0.871, 0.871$. The gate-aggregate effect $B \mid A = C = 1$ is -3.268 ± 1.365 points, negative for all three seeds $(-1.852, -4.575, -3.377)$. The coarse detector-fraction effect $C \mid A = B = 1$ is 0.036 ± 0.166 points, with one positive, one zero, and one negative paired-seed value. These results do not support a uniform benefit from the three detector-dependent pathways. The average A, B, and C effects across the other factors are 0.835 ± 2.521, -2.179 ± 1.869, and -0.472 ± 0.150 points, respectively; the full-minus-strict change is -1.816 ± 4.053 points. The complete conditional effects and interactions are retained below, including negative and near-zero outcomes.

These are retrained contrasts, so they include changes in optimization and learned parameters rather than instantaneous attention interventions. A does not remove all fine information, C tests only the coarse detector fraction, and R compares soft against constant reliability, not against a hard threshold. Three seeds characterize variation under one training recipe on one previously used validation population; they are not independent datasets. Sample standard deviations are not confidence intervals, and no significance test, winner-selection rule, or promotion of a configuration to the official test set is performed.

Availability. The original executed source, fixed protocol, and portable reproduction entry points are retained at commit `cc78751` in https://github.com/zhaiji0727/PARE, under `experiments/followup_20261103/`. The 42 evaluated fit-only heads are in `checkpoints/followup_20261103/`, and complete ranks, metrics, and all contrasts are in `benchmarks/followup_20261103/`. The earlier main-test release `6e4ed9a` and its original model definition remain unchanged in Git history. Large score matrices are retained in the complete experiment archive with published hashes; the distributed ranks, heads, and reproduction commands allow independent verification.

Complete numerical record. The following tables report all 42 runs, all 14 configuration summaries, all 36 contrast summaries, and the nine primary summaries. The accompanying CSV/JSON files additionally retain all 108 within-seed contrast values. CSV/JSON records retain unrounded values and artifact hashes; displayed values are rounded to three decimals.

Table S30: All 42 exploratory validation runs. Metrics are percentages; every query uses the full 918-image validation gallery.

Configuration	Seed	R@1	R@5	R@10	MRR@10
ABC000_R1T1	0	5.120	16.558	26.471	10.208
ABC000_R1T1	1	3.922	17.538	27.124	9.562
ABC000_R1T1	2	2.505	12.418	21.678	7.075
ABC001_R1T1	0	4.466	15.795	26.144	9.570
ABC001_R1T1	1	3.813	16.993	26.144	9.405
ABC001_R1T1	2	2.505	12.092	20.044	6.859
ABC010_R1T1	0	3.377	13.508	24.183	7.887
ABC010_R1T1	1	3.922	15.033	23.856	8.671
ABC010_R1T1	2	3.159	14.161	23.965	8.173
ABC011_R1T1	0	3.595	13.508	23.203	8.107
ABC011_R1T1	1	3.377	13.943	22.658	7.678
ABC011_R1T1	2	3.704	13.943	23.420	8.268
ABC100_R1T1	0	4.031	14.597	25.708	9.136
ABC100_R1T1	1	4.139	16.885	26.362	9.523

Configuration	Seed	R@1	R@5	R@10	MRR@10
ABC100_R1T1	2	4.684	16.776	27.669	10.216
ABC101_R1T1	0	4.031	14.488	25.599	9.089
ABC101_R1T1	1	4.139	16.885	26.362	9.523
ABC101_R1T1	2	4.684	16.776	27.669	10.216
ABC110_R1T1	0	4.357	13.617	23.529	8.662
ABC110_R1T1	1	3.595	15.033	21.786	8.408
ABC110_R1T1	2	3.268	14.706	24.401	8.332
ABC111_R1T1	0	4.357	13.617	23.747	8.682
ABC111_R1T1	1	3.595	15.033	21.786	8.429
ABC111_R1T1	2	3.268	14.815	24.292	8.277
ABC000_R0T0	0	0.654	3.486	6.318	2.115
ABC000_R0T0	1	0.545	4.031	8.388	2.270
ABC000_R0T0	2	0.436	3.377	7.190	1.970
ABC000_R0T1	0	4.793	16.885	26.906	10.042
ABC000_R0T1	1	3.704	17.974	28.540	9.976
ABC000_R0T1	2	4.031	17.865	27.560	9.555
ABC000_R1T0	0	0.871	4.248	6.536	2.123
ABC000_R1T0	1	0.545	4.466	8.170	2.338
ABC000_R1T0	2	0.763	4.357	8.388	2.529
ABC111_R0T0	0	1.743	6.100	11.002	3.865
ABC111_R0T0	1	1.634	5.991	11.220	3.785
ABC111_R0T0	2	1.198	6.209	10.675	3.409
ABC111_R0T1	0	4.466	13.834	22.440	8.654
ABC111_R0T1	1	3.268	14.706	23.094	8.215
ABC111_R0T1	2	3.704	14.488	24.183	8.538
ABC111_R1T0	0	1.852	5.991	11.874	3.997
ABC111_R1T0	1	1.416	5.773	10.675	3.608
ABC111_R1T0	2	1.416	5.991	10.675	3.441

Table S31: All 14 configurations: mean and sample standard deviation over the same three seeds (percent).

Configuration	R@1	R@5	R@10	MRR@10
ABC000_R1T1	3.849 ± 1.309	15.505 ± 2.717	25.091 ± 2.974	8.948 ± 1.654
ABC001_R1T1	3.595 ± 0.998	14.960 ± 2.555	24.110 ± 3.522	8.611 ± 1.520
ABC010_R1T1	3.486 ± 0.393	14.234 ± 0.765	24.001 ± 0.166	8.244 ± 0.397
ABC011_R1T1	3.558 ± 0.166	13.798 ± 0.252	23.094 ± 0.393	8.018 ± 0.305
ABC100_R1T1	4.285 ± 0.350	16.086 ± 1.290	26.580 ± 0.998	9.625 ± 0.547
ABC101_R1T1	4.285 ± 0.350	16.049 ± 1.353	26.543 ± 1.047	9.609 ± 0.568
ABC110_R1T1	3.740 ± 0.559	14.452 ± 0.741	23.239 ± 1.331	8.468 ± 0.173
ABC111_R1T1	3.740 ± 0.559	14.488 ± 0.763	23.275 ± 1.318	8.462 ± 0.204
ABC000_R0T0	0.545 ± 0.109	3.631 ± 0.350	7.298 ± 1.039	2.119 ± 0.150
ABC000_R0T1	4.176 ± 0.559	17.574 ± 0.600	27.669 ± 0.822	9.858 ± 0.265
ABC000_R1T0	0.726 ± 0.166	4.357 ± 0.109	7.698 ± 1.012	2.330 ± 0.203
ABC111_R0T0	1.525 ± 0.288	6.100 ± 0.109	10.966 ± 0.274	3.687 ± 0.243
ABC111_R0T1	3.813 ± 0.607	14.343 ± 0.454	23.239 ± 0.880	8.469 ± 0.228
ABC111_R1T0	1.561 ± 0.252	5.919 ± 0.126	11.075 ± 0.692	3.682 ± 0.285

Table S32: All 36 predeclared contrasts: paired-seed mean and sample standard deviation in percentage points. No significance test or winner selection is performed.

Contrast	R@1	R@5	R@10	MRR@10
A_given_B0C0	$+0.436 \pm 1.645$	$+0.581 \pm 3.335$	$+1.489 \pm 3.899$	$+0.677 \pm 2.196$
A_given_B0C1	$+0.690 \pm 1.344$	$+1.089 \pm 3.170$	$+2.433 \pm 4.513$	$+0.998 \pm 2.064$
A_given_B1C0	$+0.254 \pm 0.666$	$+0.218 \pm 0.288$	-0.763 ± 1.256	$+0.224 \pm 0.522$
A_given_B1C1	$+0.182 \pm 0.600$	$+0.690 \pm 0.515$	$+0.182 \pm 0.926$	$+0.445 \pm 0.387$
A_average_effect	$+0.390 \pm 0.535$	$+0.645 \pm 1.757$	$+0.835 \pm 2.521$	$+0.586 \pm 0.941$
B_given_A0C0	-0.363 ± 1.239	-1.271 ± 2.624	-1.089 ± 2.965	-0.705 ± 1.717
B_given_A0C1	-0.036 ± 1.091	-1.162 ± 2.638	-1.017 ± 3.815	-0.594 ± 1.739
B_given_A1C0	-0.545 ± 0.871	-1.634 ± 0.576	-3.341 ± 1.200	-1.157 ± 0.706
B_given_A1C1	-0.545 ± 0.871	-1.561 ± 0.600	-3.268 ± 1.365	-1.147 ± 0.767
B_average_effect	-0.372 ± 0.123	-1.407 ± 1.154	-2.179 ± 1.869	-0.901 ± 0.495
C_given_A0B0	-0.254 ± 0.350	-0.545 ± 0.218	-0.980 ± 0.654	-0.337 ± 0.262
C_given_A0B1	$+0.073 \pm 0.559$	-0.436 ± 0.576	-0.908 ± 0.333	-0.226 ± 0.667
C_given_A1B0	0.000 ± 0.000	-0.036 ± 0.063	-0.036 ± 0.063	-0.015 ± 0.027

Contrast	R@1	R@5	R@10	MRR@10
C_given_A1B1	0.000 ± 0.000	$+0.036 \pm 0.063$	$+0.036 \pm 0.166$	-0.005 ± 0.043
C_average_effect	-0.045 ± 0.160	-0.245 ± 0.152	-0.472 ± 0.150	-0.146 ± 0.123
interaction_AB	-0.345 ± 1.988	-0.381 ± 3.000	-2.251 ± 3.498	-0.503 ± 2.388
interaction_AC	$+0.091 \pm 0.319$	$+0.490 \pm 0.283$	$+0.944 \pm 0.206$	$+0.271 \pm 0.284$
interaction_BC	$+0.163 \pm 0.340$	$+0.091 \pm 0.354$	$+0.073 \pm 0.363$	$+0.061 \pm 0.439$
interaction_ABC	-0.327 ± 0.680	-0.036 ± 0.600	0.000 ± 1.106	-0.101 ± 0.856
full_minus_strict_RT11	-0.109 ± 0.786	-1.017 ± 2.964	-1.816 ± 4.053	-0.486 ± 1.475
ABC000_R_given_T0	$+0.182 \pm 0.166$	$+0.726 \pm 0.274$	$+0.399 \pm 0.725$	$+0.211 \pm 0.303$
ABC000_R_given_T1	-0.327 ± 1.039	-2.070 ± 2.925	-2.578 ± 2.903	-0.909 ± 1.390
ABC000_T_given_R0	$+3.631 \pm 0.491$	$+13.943 \pm 0.545$	$+20.370 \pm 0.218$	$+7.739 \pm 0.174$
ABC000_T_given_R1	$+3.123 \pm 1.272$	$+11.147 \pm 2.700$	$+17.393 \pm 3.587$	$+6.618 \pm 1.845$
ABC000_R_average_effect	-0.073 ± 0.463	-0.672 ± 1.357	-1.089 ± 1.141	-0.349 ± 0.545
ABC000_T_average_effect	$+3.377 \pm 0.768$	$+12.545 \pm 1.148$	$+18.882 \pm 1.812$	$+7.179 \pm 1.002$
ABC000_interaction_RT	-0.508 ± 1.165	-2.796 ± 3.146	-2.977 ± 3.564	-1.121 ± 1.691
ABC000_joint_RT11_minus_RT00	$+3.300 \pm 1.200$	$+11.874 \pm 2.462$	$+17.792 \pm 2.948$	$+6.830 \pm 1.546$
ABC111_R_given_T0	$+0.036 \pm 0.227$	-0.182 ± 0.063	$+0.109 \pm 0.714$	-0.005 ± 0.158
ABC111_R_given_T1	-0.073 ± 0.383	$+0.145 \pm 0.314$	$+0.036 \pm 1.309$	-0.007 ± 0.239
ABC111_T_given_R0	$+2.288 \pm 0.576$	$+8.243 \pm 0.491$	$+12.273 \pm 1.091$	$+4.782 \pm 0.350$
ABC111_T_given_R1	$+2.179 \pm 0.327$	$+8.569 \pm 0.846$	$+12.200 \pm 1.284$	$+4.781 \pm 0.083$
ABC111_R_average_effect	-0.018 ± 0.083	-0.018 ± 0.126	$+0.073 \pm 1.008$	-0.006 ± 0.099
ABC111_T_average_effect	$+2.233 \pm 0.357$	$+8.406 \pm 0.666$	$+12.237 \pm 1.151$	$+4.782 \pm 0.183$
ABC111_interaction_RT	-0.109 ± 0.607	$+0.327 \pm 0.377$	-0.073 ± 0.619	-0.002 ± 0.353
ABC111_joint_RT11_minus_RT00	$+2.215 \pm 0.350$	$+8.388 \pm 0.786$	$+12.309 \pm 1.571$	$+4.776 \pm 0.118$

Table S33: The nine predeclared primary descriptive contrasts: paired-seed mean and sample standard deviation in percentage points. No significance test or winner selection is performed.

Contrast	R@1	R@5	R@10	MRR@10
A_given_B1C1	$+0.182 \pm 0.600$	$+0.690 \pm 0.515$	$+0.182 \pm 0.926$	$+0.445 \pm 0.387$
B_given_A1C1	-0.545 ± 0.871	-1.561 ± 0.600	-3.268 ± 1.365	-1.147 ± 0.767
C_given_A1B1	0.000 ± 0.000	$+0.036 \pm 0.063$	$+0.036 \pm 0.166$	-0.005 ± 0.043
ABC000_R_given_T1	-0.327 ± 1.039	-2.070 ± 2.925	-2.578 ± 2.903	-0.909 ± 1.390
ABC000_T_given_R1	$+3.123 \pm 1.272$	$+11.147 \pm 2.700$	$+17.393 \pm 3.587$	$+6.618 \pm 1.845$
ABC000_joint_RT11_minus_RT00	$+3.304 \pm 1.200$	$+11.874 \pm 2.462$	$+17.792 \pm 2.948$	$+6.830 \pm 1.546$
ABC111_R_given_T1	-0.073 ± 0.383	$+0.145 \pm 0.314$	$+0.036 \pm 1.309$	-0.007 ± 0.239
ABC111_T_given_R1	$+2.179 \pm 0.327$	$+8.569 \pm 0.846$	$+12.200 \pm 1.284$	$+4.781 \pm 0.083$
ABC111_joint_RT11_minus_RT00	$+2.215 \pm 0.350$	$+8.388 \pm 0.786$	$+12.309 \pm 1.571$	$+4.776 \pm 0.118$

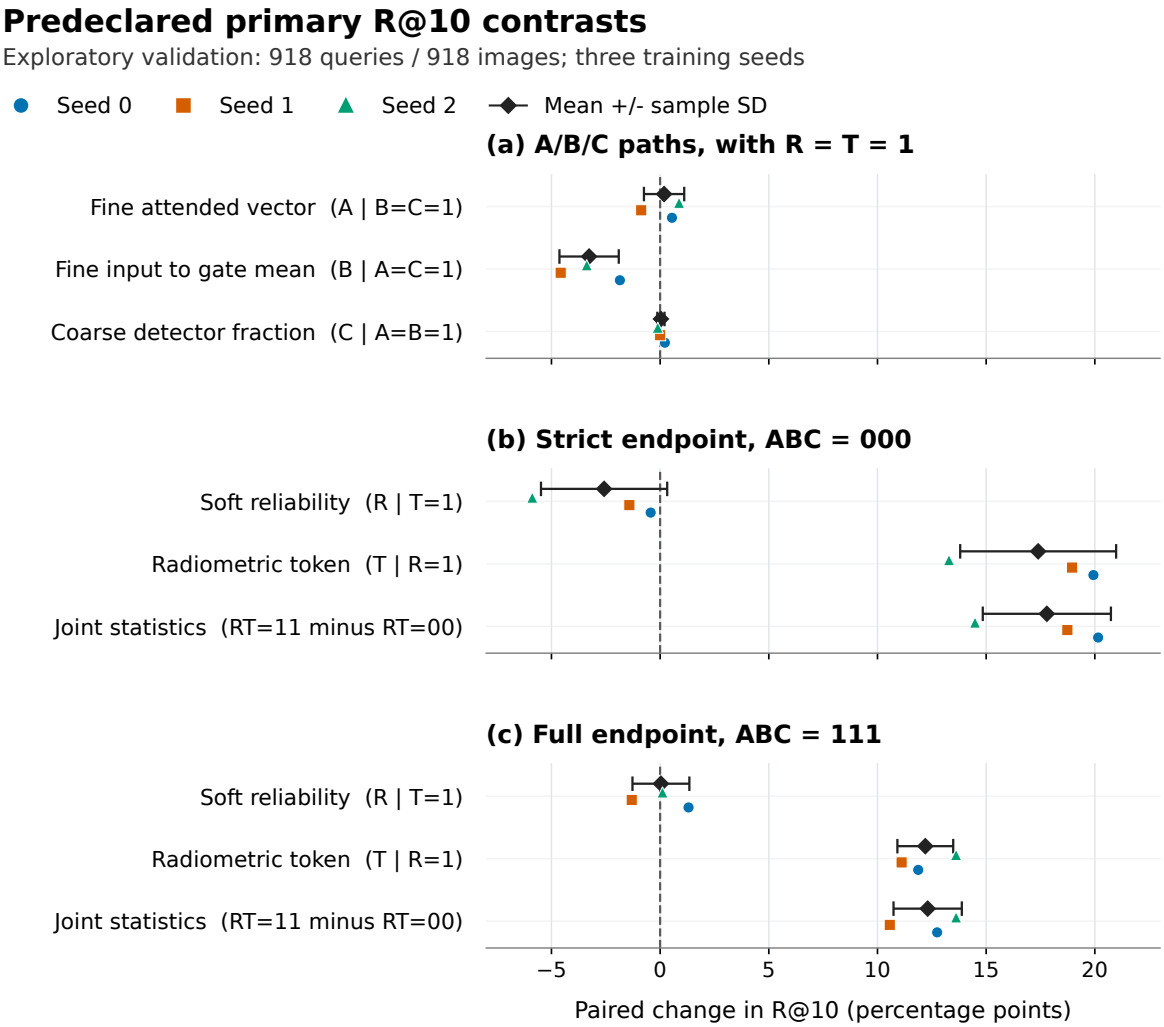


Figure S1: All nine predeclared primary R@10 contrasts, in protocol order: ABC pathways, strict-endpoint RT, then full-endpoint RT. Colored markers show individual paired-seed effects; black diamonds and horizontal error bars show their mean and sample standard deviation. Vertical offsets separate markers within each row and have no quantitative meaning. The dashed line denotes zero change. Standard deviations are not confidence intervals. This is exploratory analysis on 918 validation queries and 918 images, with three training seeds; no significance test is implied.

S9 Subsequent Conditional-Token Mechanism Diagnostic

This separate study uses ViT-B/32 with fine attention retained, fine features excluded from the gate mean, and the original coarse descriptor, reliability and token formulas. Thus it is the $A1B0C1$ configuration, not PARE-S ($A0B0C0$). We train five fixed seeds for each of three models: regional descriptors entering the token head; branchwise constant training-mean descriptors entering the same head; and token augmentation disabled. All other settings and initializations are matched. The constant means use only valid crops from the 6,610 training images (23,016 pairs): 36,887 coarse crops and 52,333 fine crops. They are not fitted to either evaluation set.

The original 918-image development set and a 1,653-image internal-check set are both previously used data. Each uses one first-caption query per image and the complete within-split gallery. The study is exploratory, not a new independent test. All 15 heads were trained for ten epochs with learning rate 0.002, contrastive temperature 0.03 and batch size 64. No configuration or seed was selected after observing these results.

Paired contrast, R@10 percentage points	Development	Internal check
Conditional minus constant descriptor	-0.3050 ± 0.2362	$+0.2178 \pm 0.5843$
Conditional minus token-off	$+12.3747 \pm 1.4216$	$+10.5989 \pm 1.2599$

Values are paired means and sample standard deviations over five seeds. For the four prespecified primary comparisons, Bonferroni family-95% t intervals are, respectively, $[-0.7607, 0.1507]$, $[-0.9097, 1.3453]$, $[9.6317, 15.1177]$, and $[8.1679, 13.0299]$ percentage points (in the table's first-row then second-row order). The intervals use four degrees of freedom and a small-sample normality approximation. Token increments are positive for every seed in both sets; conditional versus constant input does not establish a stable additional increment. This does not prove equivalence and does not identify physical scattering types. Because this experiment retains fine attention, its results cannot be transferred directly to PARE-S. All 15 heads, 30 evaluations and 105 recorded artifacts passed independent verification; the score matrices remain in the complete research archive.

S10 Matched Simplified-Model Study

S10.1 Exact configurations and retained comparator

PARE-S uses the original head with `cfar_strict` enabled. PARE-F uses no ablation flags. PARE-S without token augmentation enables `cfar_strict,ty`; it retains soft reliability. These are the only three newly trained configurations. The descriptor's fourth coordinate is fixed to zero for PARE-S, while the existing four-input network shape is retained. The original two sigmoid outputs remain. With the second output written g_b rather than the historical g_f, the simplified fusion is

$$\operatorname{norm}\big(g_c v^c + (g_b + [1 - g_c - g_b]_+)e^g\big).$$

The implementation preserves the original three-term operation order. The second output is not removed or interpreted as an active fine branch. No constant-token, reliability-off, one-gate or softmax-fusion model is introduced in this study.

For reproducibility of the PARE-F comparison, define m_{17} and m_5 as the local means on centered 17×17 and 5×5 pixel windows. At valid centers, the nonnegative background is

$$b = \max\left(\frac{289m_{17} - 25m_5}{264}, 0\right), \qquad d = \mathbf{1}[I > 2.5b + 10^{-3}].$$

The outer eight-pixel border has no valid detection center and is set false. This is a fixed CFAR-inspired multiplier, not a calibrated false-alarm probability. Window sums and moments use float64 arithmetic. Among remaining detections, choose the highest-intensity point. For an intensity tie, the first point is the one closest to the image center $((W-1)/2, (H-1)/2)$; later points maximize the minimum squared distance to already selected points. Residual ties use ascending (y, x) order. After selection, suppress candidates satisfying both $|\Delta x| \le 12$ and $|\Delta y| \le 12$. Stop after eight points or when none remain. Each selected center yields the half-open box $[x-12, x+12) \times [y-12, y+12)$, clipped to image bounds.

Fine embeddings enter the fine-attention path and the unaugmented mean of scene, valid coarse and valid fine embeddings supplied to the gate. If there is no fine crop, the fine attended vector is the scene vector. The coarse descriptor's fourth coordinate is the mean binary detection value within its crop. These are the three inputs jointly removed in PARE-S. All sizes denote stored input pixels, before per-crop encoder resizing; no 48×48 setting is used.

S10.2 Training, pairing and data history

Five frozen backbones and seeds 0–4 give 75 fresh heads. All models use the 9,181-image, 31,945-pair retained training split for ten epochs with batch size 64 (500 batches per epoch, final batch of nine), original one-direction caption-to-image cross entropy, AdamW weight decay 10^{-4} and float32 computation. Every within-backbone seed shares its initial parameters and minibatch order across the three configurations. The original validation-selected learning rate and contrastive temperature are retained for every variant; no new search or early stopping occurs. The ten seed-0 PARE-F/PARE-S models are compatibility anchors and are included in the total of 75, not counted twice.

The initial SigLIP anchor attempt stopped before training because the offline tokenizer cache directory was not set in the process environment. Pointing `HF_HOME` to the already prepared model cache resolved that setup failure. The frozen numerical sources and study protocol were unchanged; no trained head or partially optimized run was replaced. The initial failure log, cache-file hashes, recovery record and subsequent stage logs are retained in the public execution history.

Evaluation uses all 3,688 test images, each with its original first-caption query and the entire 3,688-image gallery. We retain the preceding evaluator's tie convention and bind each result to its model, input-cache and source hashes. The simplified design was motivated by prior results on this same benchmark. This is a fixed prospective computational comparison following post-hoc method development, not a previously unseen test. New training seeds do not remove that limitation.

S10.3 Complete comparisons and uncertainty

For each backbone, the two primary comparisons are PARE-S minus PARE-F and PARE-S minus PARE-S token-off. We report paired seed differences, means, sample standard deviations, ordinary 95% t intervals and Bonferroni family-95% intervals over the ten comparisons (each 99.5%, df=4). A stable positive claim requires all five R@10 differences positive, the adjusted lower endpoint above zero, and mean MRR@10

difference nonnegative. Failure does not establish zero effect or equivalence. No outcomes are excluded because of their sign.

PARE-S improves mean R@10 over PARE-F by 1.5293, 0.8514, 1.5835, 1.5130 and 0.7755 percentage points for OpenCLIP ViT-B/32, ViT-B/16, RemoteCLIP, SigLIP and GeoRSCLIP. The seed-level differences are positive in 23/25 cases; SigLIP alone meets the complete stable-positive criterion. The negative differences are RemoteCLIP seed 1 (-0.3254 points) and GeoRSCLIP seed 2 (-3.0640 points). These outcomes remain in the main summary and all paired analyses. A positive average does not establish uniform improvement, and intervals crossing zero do not prove no effect.

Token augmentation raises R@10 in every one of the 25 matched pairs; mean improvements are 7.8145, 7.3753, 4.0889, 8.0098 and 7.3536 points in the same backbone order. The four backbones other than RemoteCLIP meet the complete stable-positive criterion. RemoteCLIP's adjusted interval is $[-3.0886, 11.2665]$ points despite positive differences for all five seeds. The latter pattern is retained without selecting or dropping seeds. These ablations concern the aggregate token update, not proof of distinct physical scattering mechanisms or of an independent benefit from region-varying descriptor conditioning.

Table S34: Complete paired R@10 inference in percentage points. S, F and T0 denote PARE-S, PARE-F and PARE-S without token augmentation.

Backbone	Contrast	Mean ± SD	Ordinary 95% CI	Family-95% CI
OpenCLIP B/32	$S - F$	1.53 ± 1.14	$[0.112, 2.946]$	$[-1.327, 4.386]$
OpenCLIP B/32	$S - T0$	7.81 ± 0.90	$[6.697, 8.932]$	$[5.562, 10.068]$
OpenCLIP B/16	$S - F$	0.85 ± 0.39	$[0.371, 1.331]$	$[-0.116, 1.819]$
OpenCLIP B/16	$S - T0$	7.38 ± 0.33	$[6.964, 7.787]$	$[6.545, 8.205]$
RemoteCLIP	$S - F$	1.58 ± 2.87	$[-1.980, 5.147]$	$[-5.600, 8.767]$
RemoteCLIP	$S - T0$	4.09 ± 2.87	$[0.529, 7.649]$	$[-3.089, 11.267]$
SigLIP	$S - F$	1.51 ± 0.39	$[1.031, 1.995]$	$[0.541, 2.485]$
SigLIP	$S - T0$	8.01 ± 0.52	$[7.361, 8.659]$	$[6.702, 9.318]$
GeoRSCLIP	$S - F$	0.78 ± 2.28	$[-2.050, 3.601]$	$[-4.921, 6.472]$
GeoRSCLIP	$S - T0$	7.35 ± 2.49	$[4.257, 10.450]$	$[1.110, 13.597]$

Table S35: All paired R@10 differences by seed and mean MRR@10 differences (percentage points).

Backbone	Contrast	Seed 0	Seed 1	Seed 2	Seed 3	Seed 4	ΔMRR
OpenCLIP B/32	$S - F$	0.38	1.06	0.92	3.28	2.01	0.48
OpenCLIP B/32	$S - T0$	8.05	7.19	8.05	6.72	9.06	2.56
OpenCLIP B/16	$S - F$	1.08	0.52	0.41	1.33	0.92	0.28
OpenCLIP B/16	$S - T0$	7.35	6.83	7.43	7.70	7.57	2.53
RemoteCLIP	$S - F$	1.03	-0.33	6.64	0.19	0.38	0.69
RemoteCLIP	$S - T0$	2.87	2.33	9.19	2.96	3.09	1.42
SigLIP	$S - F$	1.22	2.06	1.25	1.79	1.25	0.73
SigLIP	$S - T0$	7.54	8.65	7.94	8.43	7.48	2.89
GeoRSCLIP	$S - F$	2.30	2.14	-3.06	2.06	0.43	0.10
GeoRSCLIP	$S - T0$	9.49	4.45	4.85	9.35	8.62	2.60

Table S36: All fixed training seeds; full-gallery percentages.

Backbone	Model	Seed	R@1	R@5	R@10	MRR@10
OpenCLIP B/32	PARE-F	0	1.06	5.10	10.01	3.03
OpenCLIP B/32	PARE-F	1	1.11	5.04	8.92	2.94
OpenCLIP B/32	PARE-F	2	1.08	4.85	9.54	2.88
OpenCLIP B/32	PARE-F	3	0.87	3.74	6.29	2.21
OpenCLIP B/32	PARE-F	4	1.41	5.18	9.06	3.15
OpenCLIP B/32	PARE-S	0	1.38	5.80	10.39	3.43
OpenCLIP B/32	PARE-S	1	1.19	5.59	9.98	3.20
OpenCLIP B/32	PARE-S	2	1.30	5.78	10.47	3.40

Table S36: All fixed training seeds; full-gallery percentages.

Backbone	Model	Seed	R@1	R@5	R@10	MRR@10
OpenCLIP B/32	PARE-S	3	1.22	5.29	9.57	3.10
OpenCLIP B/32	PARE-S	4	1.11	6.37	11.06	3.47
OpenCLIP B/32	PARE-S w/o T	0	0.19	1.19	2.33	0.66
OpenCLIP B/32	PARE-S w/o T	1	0.38	1.33	2.79	0.89
OpenCLIP B/32	PARE-S w/o T	2	0.35	1.17	2.41	0.77
OpenCLIP B/32	PARE-S w/o T	3	0.27	1.49	2.85	0.86
OpenCLIP B/32	PARE-S w/o T	4	0.24	0.98	2.01	0.62
OpenCLIP B/16	PARE-F	0	1.27	5.07	8.95	2.97
OpenCLIP B/16	PARE-F	1	1.14	5.10	9.14	2.99
OpenCLIP B/16	PARE-F	2	1.30	5.45	9.52	3.24
OpenCLIP B/16	PARE-F	3	1.11	5.45	9.08	3.03
OpenCLIP B/16	PARE-F	4	1.49	5.75	9.11	3.39
OpenCLIP B/16	PARE-S	0	1.38	5.91	10.03	3.46
OpenCLIP B/16	PARE-S	1	1.25	5.80	9.65	3.22
OpenCLIP B/16	PARE-S	2	1.49	6.13	9.92	3.47
OpenCLIP B/16	PARE-S	3	1.22	5.67	10.41	3.37
OpenCLIP B/16	PARE-S	4	1.55	6.05	10.03	3.53
OpenCLIP B/16	PARE-S w/o T	0	0.38	1.60	2.68	0.90
OpenCLIP B/16	PARE-S w/o T	1	0.38	1.65	2.82	0.92
OpenCLIP B/16	PARE-S w/o T	2	0.35	1.33	2.49	0.83
OpenCLIP B/16	PARE-S w/o T	3	0.43	1.38	2.71	0.91
OpenCLIP B/16	PARE-S w/o T	4	0.33	1.41	2.47	0.83
RemoteCLIP	PARE-F	0	0.41	2.03	3.47	1.11
RemoteCLIP	PARE-F	1	0.52	2.44	4.31	1.39
RemoteCLIP	PARE-F	2	0.35	2.14	4.12	1.15
RemoteCLIP	PARE-F	3	0.49	2.66	4.34	1.40
RemoteCLIP	PARE-F	4	0.60	2.49	4.37	1.43
RemoteCLIP	PARE-S	0	0.70	2.33	4.50	1.56
RemoteCLIP	PARE-S	1	0.73	2.22	3.99	1.49
RemoteCLIP	PARE-S	2	1.68	6.56	10.76	3.85
RemoteCLIP	PARE-S	3	0.43	2.20	4.53	1.29
RemoteCLIP	PARE-S	4	0.79	2.87	4.75	1.73
RemoteCLIP	PARE-S w/o T	0	0.27	0.79	1.63	0.56
RemoteCLIP	PARE-S w/o T	1	0.24	0.81	1.65	0.55
RemoteCLIP	PARE-S w/o T	2	0.30	0.79	1.57	0.57
RemoteCLIP	PARE-S w/o T	3	0.24	0.81	1.57	0.54
RemoteCLIP	PARE-S w/o T	4	0.30	0.81	1.65	0.58
SigLIP	PARE-F	0	1.22	4.91	8.22	2.78
SigLIP	PARE-F	1	1.03	5.07	8.62	2.82
SigLIP	PARE-F	2	1.30	5.37	8.65	3.03
SigLIP	PARE-F	3	0.98	4.66	8.51	2.73
SigLIP	PARE-F	4	1.19	4.77	8.16	2.82
SigLIP	PARE-S	0	1.52	5.45	9.44	3.27
SigLIP	PARE-S	1	1.82	6.24	10.68	3.84
SigLIP	PARE-S	2	1.55	5.69	9.90	3.49
SigLIP	PARE-S	3	1.46	6.26	10.30	3.64
SigLIP	PARE-S	4	1.79	5.64	9.41	3.60
SigLIP	PARE-S w/o T	0	0.22	1.14	1.90	0.64
SigLIP	PARE-S w/o T	1	0.30	1.27	2.03	0.72
SigLIP	PARE-S w/o T	2	0.22	1.14	1.95	0.65
SigLIP	PARE-S w/o T	3	0.35	1.17	1.87	0.74
SigLIP	PARE-S w/o T	4	0.22	1.11	1.93	0.64
GeoRSCLIP	PARE-F	0	1.49	6.02	10.03	3.56
GeoRSCLIP	PARE-F	1	0.57	2.87	5.21	1.64
GeoRSCLIP	PARE-F	2	1.95	5.99	10.79	3.96
GeoRSCLIP	PARE-F	3	1.63	6.29	10.14	3.66
GeoRSCLIP	PARE-F	4	1.82	6.34	11.09	4.02
GeoRSCLIP	PARE-S	0	1.98	7.08	12.34	4.31
GeoRSCLIP	PARE-S	1	1.14	4.47	7.35	2.63
GeoRSCLIP	PARE-S	2	0.84	4.47	7.73	2.44
GeoRSCLIP	PARE-S	3	1.63	6.81	12.20	4.01

Table S36: All fixed training seeds; full-gallery percentages.

Backbone	Model	Seed	R@1	R@5	R@10	MRR@10
GeoRSCLIP	PARE-S	4	1.74	6.59	11.52	3.96
GeoRSCLIP	PARE-S w/o T	0	0.33	1.33	2.85	0.87
GeoRSCLIP	PARE-S w/o T	1	0.30	1.38	2.90	0.86
GeoRSCLIP	PARE-S w/o T	2	0.33	1.36	2.87	0.87
GeoRSCLIP	PARE-S w/o T	3	0.33	1.36	2.85	0.86
GeoRSCLIP	PARE-S w/o T	4	0.33	1.36	2.90	0.87

Independent verification. All 75 heads, 105 model evaluations, seven scene references and ten compatibility anchors passed the independent numerical audit. The ten anchors exactly matched prior parameters, per-epoch losses, complete score matrices and ranks. The auditor checked full matrices, not only reported summary metrics. A separate statistics implementation reproduced all 15 model summaries, ten primary contrasts, nine scale summaries and twelve observational-group summaries without selecting outcomes. The frozen-source manifest has SHA-256 prefix `fb39609b2d5c`; the independent numerical audit is `be6ffe2ef68d`, and the statistics proof is `be948befa4da`. Complete hashes and exact per-result model/input bindings accompany release `b174e52fa987085f09b2c75ca173fa0ca042c484` at https://github.com/zhaiji0727/PARE. Public copies of the 75 heads redact only local-path metadata; the projection manifest records original/public hashes and verifies every parameter's data type, shape and bytes. The supplied portable scorer was additionally checked on the six seed-0 ViT-B/32 and SigLIP heads: all complete score matrices and ranks matched exactly (maximum score difference zero). This checks the scoring entry point; it does not claim that the portable training entry point was used for the recorded 75 runs.

S10.4 New-model scale and observational-group assessment

For ViT-B/32, all three variants and five seeds are evaluated at both 1.5× and 2× synthetic pixel-spacing factors, yielding 30 additional model evaluations. Original-scale results are the same records as the main evaluation. The existing transformation converts to RGB, resizes with Lanczos to rounded width/factor and height/factor, does not resize back to the input raster, and then applies the unchanged extraction and per-crop encoder preprocessing. Model parameters are fixed across scales. These are synthetic sampling changes, not real sensor-GSD changes.

The fixed metadata groups use each stored axis rounded to 0.01 m with round-half-up, requiring agreement between the rounded axes. The groups are below 0.30 m/pixel (760 queries), 0.30 to below 0.45 (238), at least 0.45 (188), and unavailable (2,502). Every grouped rank is drawn from the unchanged complete gallery. Geography and scene composition can confound these observational results.

The table below reports mean R@10 and sample SD. PARE-S retains higher mean R@10 than PARE-F and token-off at all three scales. It loses 1.3124 ± 0.4548 points at factor 1.5 and 1.8492 ± 0.2264 points at factor 2 relative to its own original-scale result. The corresponding PARE-F decreases are 0.8948 ± 0.4058 and 2.2451 ± 0.9958 points. Thus simplification does not uniformly reduce the scale-induced loss. The absolute accuracy advantage and sensitivity to resampling are distinct observations. These secondary scale comparisons are descriptive; they do not receive the primary ten-comparison inferential claim.

Factor	PARE-F	PARE-S w/o T	PARE-S
1	8.76 ± 1.45	2.48 ± 0.35	10.29 ± 0.56
1.5	7.87 ± 1.24	2.34 ± 0.28	8.98 ± 0.58
2	6.52 ± 1.08	2.42 ± 0.24	8.44 ± 0.42

Table S37: ViT-B/32 scale changes relative to each seed's original-scale result, in percentage points. Means and sample SDs are descriptive.

Factor	Model	ΔR@1	ΔR@5	ΔR@10	ΔMRR@10
1.5	PARE-F	-0.01 ± 0.23	-0.43 ± 0.13	-0.89 ± 0.41	-0.21 ± 0.14
1.5	PARE-S	-0.20 ± 0.08	-0.84 ± 0.27	-1.31 ± 0.45	-0.48 ± 0.11
1.5	PARE-S w/o T	-0.02 ± 0.08	0.00 ± 0.10	-0.14 ± 0.26	-0.03 ± 0.07
2	PARE-F	-0.44 ± 0.19	-1.07 ± 0.61	-2.25 ± 1.00	-0.85 ± 0.32
2	PARE-S	-0.27 ± 0.10	-1.10 ± 0.33	-1.85 ± 0.23	-0.66 ± 0.13

Table S37: ViT-B/32 scale changes relative to each seed's original-scale result, in percentage points. Means and sample SDs are descriptive.

Factor	Model	ΔR@1	ΔR@5	ΔR@10	ΔMRR@10
2	PARE-S w/o T	0.01 ± 0.11	0.24 ± 0.12	-0.05 ± 0.18	0.04 ± 0.08

Table S38: ViT-B/32 scale evaluations for all fixed seeds; full-gallery percentages.

Scale	Model	Seed	R@1	R@5	R@10	MRR@10
1	PARE-F	0	1.06	5.10	10.01	3.03
1	PARE-F	1	1.11	5.04	8.92	2.94
1	PARE-F	2	1.08	4.85	9.54	2.88
1	PARE-F	3	0.87	3.74	6.29	2.21
1	PARE-F	4	1.41	5.18	9.06	3.15
1	PARE-S	0	1.38	5.80	10.39	3.43
1	PARE-S	1	1.19	5.59	9.98	3.20
1	PARE-S	2	1.30	5.78	10.47	3.40
1	PARE-S	3	1.22	5.29	9.57	3.10
1	PARE-S	4	1.11	6.37	11.06	3.47
1	PARE-S w/o T	0	0.19	1.19	2.33	0.66
1	PARE-S w/o T	1	0.38	1.33	2.79	0.89
1	PARE-S w/o T	2	0.35	1.17	2.41	0.77
1	PARE-S w/o T	3	0.27	1.49	2.85	0.86
1	PARE-S w/o T	4	0.24	0.98	2.01	0.62
1.5	PARE-F	0	1.36	4.56	8.41	2.88
1.5	PARE-F	1	0.89	4.80	8.30	2.63
1.5	PARE-F	2	1.19	4.50	8.70	2.85
1.5	PARE-F	3	0.62	3.25	5.67	1.82
1.5	PARE-F	4	1.41	4.66	8.27	3.00
1.5	PARE-S	0	1.25	4.91	9.35	3.03
1.5	PARE-S	1	0.98	5.04	9.33	2.85
1.5	PARE-S	2	0.98	5.02	8.84	2.84
1.5	PARE-S	3	1.03	4.56	8.03	2.65
1.5	PARE-S	4	1.00	5.10	9.35	2.84
1.5	PARE-S w/o T	0	0.27	1.03	2.28	0.70
1.5	PARE-S w/o T	1	0.27	1.36	2.20	0.74
1.5	PARE-S w/o T	2	0.27	1.27	2.33	0.72
1.5	PARE-S w/o T	3	0.30	1.55	2.82	0.85
1.5	PARE-S w/o T	4	0.24	0.95	2.09	0.62
2	PARE-F	0	0.87	4.07	7.40	2.29
2	PARE-F	1	0.49	2.93	5.29	1.55
2	PARE-F	2	0.68	4.28	7.43	2.23
2	PARE-F	3	0.52	2.96	5.40	1.61
2	PARE-F	4	0.76	4.34	7.08	2.27
2	PARE-S	0	1.08	4.66	8.60	2.80
2	PARE-S	1	1.00	4.96	8.43	2.69
2	PARE-S	2	0.89	4.72	8.41	2.61
2	PARE-S	3	1.06	4.15	7.81	2.53
2	PARE-S	4	0.84	4.83	8.98	2.68
2	PARE-S w/o T	0	0.35	1.30	2.41	0.81
2	PARE-S w/o T	1	0.27	1.60	2.44	0.83
2	PARE-S w/o T	2	0.27	1.57	2.47	0.81
2	PARE-S w/o T	3	0.30	1.63	2.74	0.88
2	PARE-S w/o T	4	0.30	1.25	2.06	0.69

Table S39: ViT-B/32 observational query groups; unchanged complete gallery, means and sample SDs over seeds.

Group (n)	Model	R@1	R@5	R@10	MRR@10
< 0.30 (760)	PARE-F	1.24 ± 0.15	4.61 ± 0.74	8.61 ± 1.55	2.88 ± 0.39
< 0.30 (760)	PARE-S	1.05 ± 0.16	5.79 ± 0.73	10.76 ± 0.52	3.18 ± 0.19
< 0.30 (760)	PARE-S w/o T	0.24 ± 0.11	0.66 ± 0.26	1.58 ± 0.42	0.51 ± 0.16
$[0.30, 0.45)$ (238)	PARE-F	0.59 ± 0.38	3.70 ± 0.69	6.97 ± 0.48	2.09 ± 0.33
$[0.30, 0.45)$ (238)	PARE-S	1.85 ± 0.82	5.38 ± 1.13	8.82 ± 1.65	3.48 ± 0.90
$[0.30, 0.45)$ (238)	PARE-S w/o T	0.17 ± 0.23	1.01 ± 0.38	1.76 ± 0.35	0.54 ± 0.12
≥ 0.45 (188)	PARE-F	3.51 ± 0.81	10.96 ± 1.44	15.96 ± 3.80	6.66 ± 1.20
≥ 0.45 (188)	PARE-S	3.94 ± 1.44	10.85 ± 1.79	17.02 ± 1.46	7.32 ± 1.13
≥ 0.45 (188)	PARE-S w/o T	0.64 ± 0.24	2.77 ± 0.95	5.43 ± 1.02	1.74 ± 0.38
Unknown (2502)	PARE-F	0.94 ± 0.20	4.48 ± 0.62	8.44 ± 1.44	2.62 ± 0.34
Unknown (2502)	PARE-S	1.04 ± 0.15	5.41 ± 0.48	9.78 ± 0.91	3.05 ± 0.28
Unknown (2502)	PARE-S w/o T	0.29 ± 0.09	1.31 ± 0.17	2.60 ± 0.37	0.78 ± 0.13

PARE-S's mean R@10 exceeds both controls within all four metadata groups. Its values of 10.7632, 8.8235, 17.0213 and 9.7842% are not monotonic with the recorded spacing. The larger value in the highest-spacing group cannot be attributed to resolution: the groups differ in scene content and provenance, and their sample sizes are unequal. Missing metadata is reported as a separate group rather than imputed.